📘 MODULE 1: SPLUNK FUNDAMENTALS (Notes)

1.1 What is Splunk?

Splunk is a platform that collects, indexes, and searches machine data (logs, metrics, events) from any source. SIEM is one use case, built on top of it with Splunk Enterprise Security.

Why it exists: Logs are scattered across thousands of machines in different formats. Searching each by hand with grep does not scale. Splunk centralizes them and lets you search everything in seconds.

Splunk vs traditional log searching

Traditional (grep on servers)	Splunk
Log in to each machine	One search interface
Different formats, manual parsing	Extracts fields automatically
No history across machines	Time-based search across all data
No alerts or dashboards	Built-in alerts, dashboards, reports

Main use cases: security monitoring (SIEM), IT operations and troubleshooting, application monitoring, compliance and audit reporting.

1.2 Splunk product editions (do not mix these up)

Product	What it is
Splunk Enterprise	Self-hosted software you install (your lab)
Splunk Cloud Platform	Splunk-managed, hosted version
Splunk Enterprise Security (ES)	A premium app on top of Splunk that adds SOC features (notable events, risk, incident review)
Splunk SOAR	A separate product for automating response (playbooks)
Splunk Free	Limited, single-user edition (about 500 MB/day indexing, no authentication/roles, and reduced features such as no distributed search). Verify exact limits in current Splunk docs.

Product	What it is
Enterprise trial	Time-limited full-featured trial (historically 60 days, 500 MB/day). Check the current download page for terms.

1.3 Architecture components

DATA SOURCE → FORWARDER → INDEXER → INDEX → SEARCH HEAD → SEARCH → DASHBOARD / ALERT

Component	Job	Analogy
Universal Forwarder (UF)	Lightweight agent. Reads data and sends it. Does almost no processing.	Delivery truck
Heavy Forwarder (HF)	Full Splunk instance that can parse, filter, mask, and route data before sending.	Sorting depot
Indexer	Receives data, parses it, stores it in indexes, and answers search requests.	Warehouse
Index	Storage location for events.	Shelf
Search Head	Web UI where users search and build dashboards/alerts.	Counter
Deployment Server	Centrally pushes config and apps to many forwarders.	HQ sending instructions
License Manager	Tracks daily indexed volume against your license.	Meter

UF vs HF (very common interview question)

	UF	HF
Size	Tiny, low CPU/RAM	Full Splunk
Parsing	No (mostly)	Yes
Web UI	No	Yes
Use	Collect from endpoints/servers	Filter, mask, route, collect from APIs

1.4 Licensing in one paragraph

Splunk licenses by **daily volume of data indexed** (GB/day), not by users or searches. Exceeding it creates license warnings, and repeated violations can restrict searching on some license types. Admins monitor this regularly.

1.5 Your lab

On a single machine, **one Splunk Enterprise install plays Search Head, Indexer, and License Manager together.** This is called a **standalone** deployment.

1.6 Hands-on (still pending from you)

Run these and send me the output:

bash

sudo /opt/splunk/bin/splunk status

- /opt/splunk/bin/splunk is the Splunk CLI. status checks if it is running.
- Expected: splunkd is running (PID: ####).
- Errors: "No such file or directory" means Splunk is installed elsewhere. "not running" means start it with … splunk start.

Then in Splunk Web (http://localhost:8000), search with time set to **All time**:

spl

index=* | head 10

Tell me what appeared and the values of host, source, sourcetype.

1.7 Troubleshooting

Symptom	Likely cause
Cannot open localhost:8000	Splunk not running, or web port is different
Search returns nothing	No data ingested yet, wrong time range, or wrong index

1.8 Interview Q&A (copy into notes)

Q: What is Splunk?
A: A platform for collecting, indexing, and searching machine-generated data. In a SOC it's used as a SIEM to detect and investigate security events.

Q: Explain Splunk architecture.
A: Forwarders collect data and send it to indexers, which parse and store it in indexes. Search heads let users search the indexers and build dashboards and alerts.

Q: UF vs HF?
A: The UF is a lightweight forwarder that only reads and sends data. The HF is a full Splunk instance that can parse, filter, mask, and route before forwarding.

Q: What does the Deployment Server do?
A: It centrally manages configurations and apps for many forwarders.

Q: How is Splunk licensed?
A: By the volume of data indexed per day.

Q: Enterprise vs Enterprise Security?
A: Enterprise is the core platform. ES is an add-on app that provides SIEM features such as notable events, risk-based alerting, and incident review.

1.9 Revision

- Splunk = collect, index, search machine data.
- Forwarder sends, Indexer stores, Search Head searches.
- UF is light, HF can parse and filter.
- License is by daily ingest volume.
- ES and SOAR are separate products from Enterprise.